

3. Row 2 Column 1 = Row 3 Column 1 = 0

Row 3 Column 2 = 0

So, the condition that for each pivot column, the elements below the pivot position must be all 0 is True.

∴ The augmented matrix $\begin{bmatrix} 1 & -3 & -4 & b_1 \\ 0 & -7 & -6 & 3b_1+b_2 \\ 0 & 0 & 0 & b_1+2b_2+b_3 \end{bmatrix}$ is in echelon form.

There is a row in the matrix $\begin{bmatrix} 0 & 0 & 0 & b_1+2b_2+b_3 \end{bmatrix}$. By Theorem 2, the system of equations corresponding to the augmented matrix $\begin{bmatrix} 1 & -3 & -4 & b_1 \\ 0 & -7 & -6 & 3b_1+b_2 \\ 0 & 0 & 0 & b_1+2b_2+b_3 \end{bmatrix}$ is consistent, i.e. there exists a solution set iff $b_1+2b_2+b_3=0$.

$b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. So $Ax=b$ has a solution when $b_1+2b_2+b_3=0$.
 $Ax=b$ has no solution when $b_1+2b_2+b_3 \neq 0$.

Exercises 17-20 refer to the matrices A and B below. Make appropriate calculations that justify your answers and mention an appropriate theorem.

$$A = \begin{bmatrix} 1 & 3 & 0 & 3 \\ -1 & -1 & -1 & 1 \\ 0 & -4 & 2 & -8 \\ 2 & 0 & 3 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 3 & -2 & 2 \\ 0 & 1 & 1 & -5 \\ 1 & 2 & -3 & 7 \\ -2 & -8 & 2 & -1 \end{bmatrix}$$

17. How many rows of A contain a pivot position? Does the equation $Ax=b$ have a solution for each b in $\mathbb{R}^4$?
 By defn, a pivot position of a matrix corresponds to the leading 1 of a row of the matrix in reduced row echelon form.

Solution:

$$\begin{bmatrix} 1 & 3 & 0 & 3 \\ -1 & -1 & -1 & 1 \\ 0 & -4 & 2 & -8 \\ 2 & 0 & 3 & -1 \end{bmatrix} \xrightarrow[\text{+ Row 1}]{\text{Row 2 = Row 2}} \begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 2 & -1 & 4 \\ 0 & -4 & 2 & -8 \\ 2 & 0 & 3 & -1 \end{bmatrix} \xrightarrow[\text{-2 \times Row 1}]{\text{Row 4 = Row 4}} \begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 2 & -1 & 4 \\ 0 & -4 & 2 & -8 \\ 0 & -6 & 3 & -7 \end{bmatrix} \xrightarrow[\text{+2 \times Row 2}]{\text{Row 3 = Row 3}} \begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 2 & -1 & 4 \\ 0 & 0 & 0 & 5 \\ 0 & -6 & 3 & -7 \end{bmatrix} \xrightarrow[\text{Row 2 and Row 3}]{\text{Interchange}} \begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 0 & 0 & 5 \\ 0 & 2 & -1 & 4 \\ 0 & -6 & 3 & -7 \end{bmatrix}$$